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A MIDDLE-TRIASSIC PALEOMAGNETIC POLE FOR THE EURASIAN PLATE FROM HEMING (FRANCE)

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Abstract. The paleomagnetic study of Middle Triassic limestones (91 cores, 160 samples) from a 40 m thick section in Heming (France) yields a paleomagnetic pole for stable Europe at lat=53.5°N, lon=140.8°E, d_p=2.3°, d_m=3.9°. This pole fills an important gap of the apparent polar wander path (APWP) of the Eurasian plate. The comparison of this pole with available data shows a standstill of the Eurasian plate from the Late Paleozoic to the Middle Triassic, followed by a rapid drift until the Lower Jurassic. The beginning of the rapid drift is contemporaneous with the initiation of main rifting processes in the Tethyan Oceans.

Introduction

Despite recent efforts from the paleomagnetic community, APWPs for major blocks are still defined to first order only [Westphal et al., 1986; Van der Voo, 1990; Besse and Courtillot, 1991]. In order to determine a new paleomagnetic pole of Middle-Triassic age, samples were collected from the Heming quarry (48.70°N, 6.97°E), located in the eastern part of the Paris basin (France). The section corresponds in age to the Upper Muschelkalk of the German basin facies. The section is approximately 40 m thick. The lower part (1-11.5 m) consists of Entroque limestones which are very massive at the base, and passes towards the summit to alternating thick limestone beds and thin marly layers. In the upper part (11.5-42 m), the Ceratite limestones consist of whitish decimeter thick limestone beds separated by marls. Ceratite zonation [Duringer and Hagdorn, 1987] gives an Illyrian age for the base of the section (0-9 m), and Fassanian for the upper part (uppermost part of the Anisian and lowermost part of the Ladinian). Forty levels distributed all over the quarry were sampled (two to nine cores per level). As usual in the Paris basin, the bedding attitude is flat, the average dips being about 6°, and will not allow a fold test.

Paleomagnetic Analysis

Paleomagnetic analyses have been conducted on 160 samples (91 cores), using a CTF 3-axis SQUID magnetometer at the Institut de Physique du Globe de Paris. This magnetometer has a noise level sufficiently low (5 10^{-3} mA/m) to allow the measurement of most of these weakly magnetized samples (2 to 22 10^{-2} mA/m before demagnetization). The monitoring of chemical transformations due to heating is done for each step using a Molyneux susceptibility bridge. An isothermal remanent magnetization (IRM) is given to sample T02 up to a maximum field of 1.2 T (Fig. 1a). Early saturation of soft coercive minerals for this sample is reached in a field <0.2T, while a fraction is clearly not saturated in progressively higher fields. In order to better characterize the magnetic carriers, we have thermally demagnetized a composite IRM in a field free oven (B<20 nT) as described in Lowrie [1990]. Different DC fields (0.12, 0.4 and 1.2 T) are applied successively to one of the three orthogonal axis of a sample which is then thermally demagnetized (Fig. 1b). The demagnetization curve shows a small contribution of a hard coercivity mineral which looses magnetization above 120°C. It probably corresponds to the presence of goethite. This mineral is found all along the section and is mainly concentrated in the samples situated closer to the surface. The high coercivity component contribution above 120°C is small. A low coercivity mineral is dominant and the intensity of magnetization gently decreases until around 575°C. It probably corresponds to a mineral of the magnetite family. Another mineral of medium coercive field is also observed, with a strong decay between 120°C and 320°-360°C. The natural remanent magnetization (NRM) curve obtained during thermal demagnetization for the sample H24C (Fig. 1c), shows a rapid decay of intensity followed by

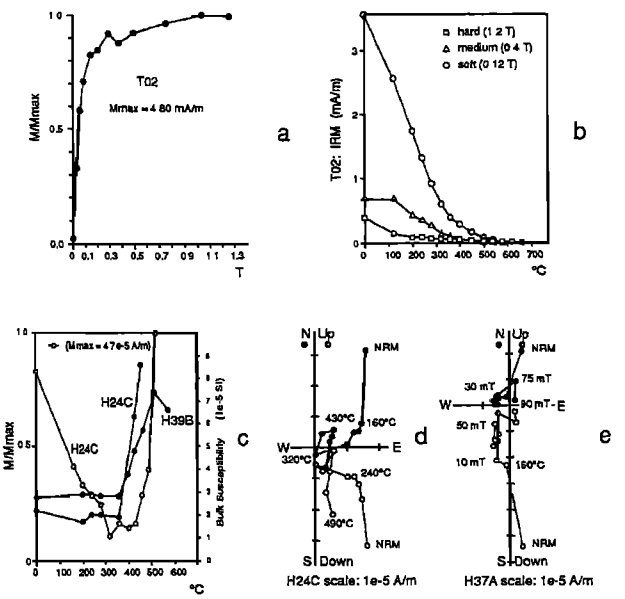


Fig. 1a. IRM acquisition curve; Fig. 1b thermal demagnetization of composite IRM; Fig. 1c, NRM decay curve (line with open circles) and evolution of low field susceptibility with temperature (lines with filled circles); Orthogonal projections of representative thermal demagnetization (Fig. 1d), alternating field demagnetization after a first heating step at 160°C (Fig. 1e). The open (filled) symbols refers to the horizontal (E-W vertical) plane. All projections are in stratigraphic coordinates.

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a significant increase above 320°C. More than 80% of the magnetization is destroyed at 320°C. A clear middle temperature component of normal polarity (MTC1) in the range 200-320°C (Fig. 1d) can be seen on the vectorial diagram. Most of the time, the path becomes erratic on the Zijderveld plot [1967] above 320°C, and no clear direction can be isolated. Alternating field demagnetizations are generally ineffective even after a preliminary thermal demagnetization step at 160°C (Fig. 1e). A significant increase of the susceptibility (χ_o) occurs above 350°C for sample H24C, indicating a sudden mineralogical change from a low χ_o to higher χ_o mineral. The susceptibility has also been measured at higher temperature for sample H39B (Fig. 1c) and shows that the increase above 350°C is followed by a decrease at 520°C. The susceptibility changes described for samples H24C and H39B along with the other characteristics described above, would suggest the presence of minerals of the iron sulfide family. In this hypothesis, the increase in susceptibility could be due to a transformation of iron sulfide into magnetite during thermal treatment, the newly created magnetite being transformed into hematite at higher temperature.

A different magnetic behaviour is observed in sample T04 (Fig. 2a). The acquisition of progressive IRM clearly saturates in a field<0.2T for some minerals, while another fraction is clearly not saturated in progressively higher fields. Thermal demagnetization of the IRM (Fig. 2b) shows an important hard coercivity contributor destroyed at 120°C corresponding to goethite. The contribution above 120°C of the hard and middle coercivity mineral is negligible (although a minor fraction of hematite with an unblocking temperature well over 600°C can be seen in the hard component). The magnetization above 120°C is dominated by a low coercivity mineral, and the intensity decreases regularly until 575°C. It also corresponds to a mineral of the magnetite family. Susceptibility curves of samples H28B and T04 (Fig. 2c) remain stable, with only a slight increase above 500°C. The natural remanent magnetization (NRM) curves obtained during thermal demagnetization for samples H28B and H29B (Fig. 2c) decrease quite regularly until around 520°C. One or two components can be seen on the Zijderveld plot during thermal demagnetization: A first component is a middle temperature component of normal polarity (MTC 2) determined between 200 and 320°C (Fig. 2d). A second component (HTC) of both polarities is determined after 320°C (Fig. 2d and 2e). Alternating field demagnetizations (after a first 160°C heating step) is also effective, more than 90% of the magnetization being destroyed at 60 mT (Fig. 2f). However, we notice that the initial susceptibility intensities shown in figure 2c are lower than those shown in figure 1c. An alternative interpretation would be that higher susceptibilities found for samples of the first behaviour correspond to the presence of paramagnetic minerals. The medium coercivity and medium blocking temperature component in figure 1b would correspond to magnetite and the increase in susceptibility could be due to the crystallization of new magnetic minerals from the paramagnetic fraction.

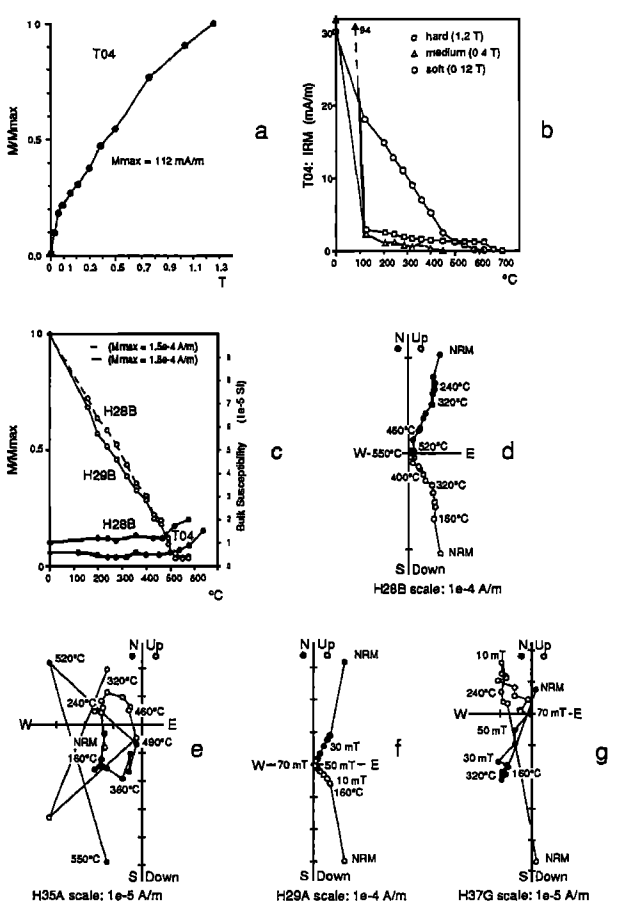


Fig. 2. Same as Figure 1 for a, b, c, d; Orthogonal projections of a thermal (Fig. 2e), alternating field (Fig. 2f) and composite thermal+alternating field (Fig. 2g) demagnetizations in stratigraphic coordinates.

The magnetization in the section is thus caused by several contributors. The goethite, present all along the section, is easily removed by heating at 120°C. In most of samples from the bottom of the section, the presence of iron sulfides or paramagnetic minerals is observed. Samples are most of the time remagnetized during heating above 350°C. A clear component of middle temperature (MTC1 in Table 1) can usually be determined by principal component analysis between 200°C and 320°C. This component is always of normal polarity. The determination of a higher temperature component is precluded by mineralogical transformations. Other samples are dominated by magnetite. The origin of the secondary component of normal polarity found in some samples (MTC2 in Table 1) is probably due to a variable contribution of iron sulfides or low blocking temperature magnetite. In general, the alternating field demagnetization leads to less scattered final directions on the Zijderveld plot. For 115 out of 160 samples, a mixed demagnetization procedure has been performed. Samples are heated up to 320°C and then demagnetized by alternating fields up to 70 mT. The result of such a procedure is shown in figure 2g for a weakly magnetized reversed sample (compare with Fig.2e). The final component most generally determines a linear segment passing through the origin. In some cases, the final intensities are close to the magnetometer noise level and highly scattered directions are observed on the Zijderveld plot. For 8 samples where the final vector trajectory hardly could be recognized, we have used the great circle intersection method of McFadden and McElhinny [1988], which combine both great circle and directions in a single mean. Both normal and reversed polarity directions are found (15% of samples are of reversed polarity). Reversed polarities are defined in four levels in the top of the section (approximately 15 m in thickness) separated by one level of normal polarity. Normal (D_S=28.4,I_S=35.3, K_S= 93.6, α_{95}=4, N=15) and reversed directions (D_S=202.4, I_S=-34.1, K_S=124.4, α_{95}=8.3, N=4) are antipodal (Fig. 3a). A reversal test [McFadden and Mc Elhinny, 1990] has been performed at the sample level and at the site level. Both are positive at the 95% level (angles of 4.4° and 5.1° respectively to compare with respective critical values at the 95% level of 8.4° and 8.4° for 58 samples or 19 sites). Since we have obtained a clear component of both

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Table 1. Mean directions for the main components before (g) and after (s) bedding correction. N is the number of levels; n, number of samples (data used are those without brackets); D and I respectively declination and inclination; k, precision parameter [Fisher, 1953]; a_{95}, confidence interval at the 95% level.HTCs, mean of samples, HTCl mean of levels. Mean virtual geomagnetic poles from the data after (s) bedding correction ; λ, latitude of the pole; ϕ, longitude; d_p,d_m, semi axis of the oval of 95% confidence.

	N	n	D_g	I_g	k_g	a_{95}	D_s	I_s	k_s	a_{95}	λ (°N)	ϕ (°E)	d_p	d_m
MTC1	22	(81)	17.8	57.2	22.4	6.7	18.1	57.1	22.5	6.7				
MTC2	16	(63)	06.6	58.6	41.2	5.8	07.2	57.9	48.0	5.4				
MTC1+2	38	(144)	13.2	57.9	27.3	4.5	13.5	57.6	28.6	4.4	75.7	139.0	4.7	6.5
HTCs	(19)	58	27.2	37.2	35.0	3.2	26.7	36.3	32.5	3.3	54.3	140.6	2.2	3.8
HTCl	19	(58)	27.5	36.0	95.6	3.4	27.1	35.1	96.9	3.4	53.5	140.8	2.3	3.9

polarities, passing through the origin, carried by a mineral of the magnetite family and a positive reversal test at the 95% level, we think that the component so far isolated is likely to be of primary origin. Mean directions are computed using Fisher's statistics [1953] for all samples in a given stratigraphic layer (Fig. 3a), considered to correspond to a site. The mean directions for the different components are summarized in Table1. The high rejection rate due to the strong laboratory remagnetizations after 350°C gives a coarse time resolution. It is thus not possible to present the results in terms of a reliable magnetostratigraphy. However, we notice that horizons of reversed polarity have also been found in various series of identical age [for a synthesis, see Molina-Garza et al, 1991].

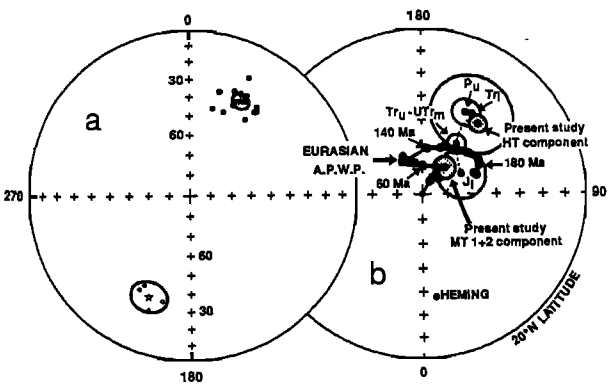


Fig. 3. (a) Equal area projection (in stratigraphic coordinates) showing the primary (HTl component) directions of both normal (filled circles) and reversed polarities (open circles) averaged at the site level (see text). Filled star, mean of levels of normal polarity; open stars, mean of levels of reversed polarity. The circles represent the uncertainties at the 95% level of confidence for normal and reversed mean directions. The filled (open) symbols refer to a projection in the lower (upper) hemisphere. (b) Polar projection showing the mean poles derived from this study with their confidence interval (grayed ellipses): MT1+2 is probably a remagnetization acquired during the lower Tertiary while HT is thought to represent a middle Triassic direction. In addition the mean upper Permian (P_u), lower Triassic (Trl), mean lower Jurassic (J_l) poles derived from our compilation (see text), the middle-upper Triassic pole (Tr_u-UTr_m) from Gallet [1988] and the Eurasian synthetic apparent polar wandering path from Besse and Courtillot [1991] for the last 200 my have been plotted as well.

Discussion

The mean directions of the MT 1 and 2 components are identical at the 95% level of confidence. We have thus computed a mean pole at the site level including MTC1 and MTC2 (Table 1) (Fig. 3b). When compared with the Cenozoic APWP for Eurasia [Besse and Courtillot, 1991], this pole is situated close to the Oligocene mean pole (Fig. 3b). MTC directions could of course result from overlapping blocking spectra between the present earth field and HTC. However the linear character of the MTC component on the Zijderveld plots favours the interpretation of this direction as a remagnetization acquired during the early-middle Cenozoic.This period is also a time of major activity in the opening of the Rhone/Rhine graben associated with an increase of local heat flux and volcanism. The mean pole computed from the high temperature component directions (Table 1) is compared with available data (Fig. 3b). An Upper Permian mean pole (P_u) can be derived from five studies [Zijderveld, 1975; Bogdanoff and Schott, 1977; Van den Ende, 1977; Merabet and Daly, 1986, and Tucholka and Gallet, 1991]. Four paleomagnetic poles [Rother, 1970; Mulder, 1972; Biquand, 1977 and Turner and Ixer, 1977] can be used to compute a mean Lower Triassic pole (Trl). This pole is ill defined (Fig. 3b) with an A_{95} of 16°. The large dispersion is due either to an insufficient number of sites (or samples) or to inadequate demagnetization by alternating field only on red sandstones in two studies. A middle-upper Triassic pole (Tr_u-Utr_m) [Gallet, 1988] is derived from the magnetostratigraphic study of in-situ oriented cores recovered from a borehole in the Paris basin spanning the middle-upper Triassic (but mainly the upper Triassic). We have added for completeness a mean Lower Jurassic pole (J_l) [Galbrun and Daly, 1987] computed from three studies [Hijab and Tarling, 1982; Fabre, 1986 and Galbrun and Daly, 1987]. The Upper Permian and Lower Triassic pole positions are indistinct at the 95% level. Our pole is situated very close to the two previous poles, the limits of its 95% interval of confidence circle intersecting those of the Upper Permian one. It is statistically distinct from the upper middle/upper Triassic pole of Gallet [1988] , and situated far away from the Lower Jurassic mean pole. Although further studies would clearly be needed to confirm our result and, more generally, the entire set of Eurasian paleopoles for the Triassic and Jurassic, it seems that Upper Permian to Middle Triassic times were a period of standstill for the Eurasian plate whereas the main drift occurred between the Ladinian and the Lower Jurassic. We note in passing that sedimentary, tectonic and magmatic evidence shows that major breakup and extension also took place in the middle Triassic and continued into the upper Triassic in the western Tethys from the Taurus to Oman

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regions [Argyriadis et al, 1980; Marcoux, 1987]. This rifting led to the opening of the Neotethys.

Conclusions

The paleomagnetic study of Aniso-Ladinian limestones in the eastern part of France provides the following results: a low blocking temperature component, carried either by iron sulfides or magnetite is interpreted as a remagnetization acquired during the genesis of Rhone/Rhine grabens during the Oligocene. Despite a very low initial intensity of magnetization, a clear component of both polarities carried by magnetite is isolated by heating and alternating field. The derived Middle Triassic pole is situated close to published Eurasian Upper Permian/Lower Triassic poles, but is remote from Upper Triassic/Lower Jurassic poles. The major drift of the Eurasian plate may likely have occurred between Middle Triassic and Lower Jurassic times. This result is consistent with the timing of the main rifting episodes in the Western Tethys. Finally, we emphasize the need for new paleopoles for the Eurasian plate during most of Mesozoic times.

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References

Argyriadis, I., Graciansky, P.C. DE, Marcoux, J.& Ricou, L.E., The opening of the Mesozoic Tethys between Eurasia and Arabia-Africa. Proc. 26th Int. Geol. Congr. Paris, C5, 199-214, 1980.

Besse, J. and Courtillot, V., Revised and synthetic apparent polar wander paths of the African, Eurasian, North American and Indian plates, and true polar wander Since 200 Ma, J. Geoph. Res., 96, 4029-4050, 1991.

Biquand, D., Paléomagnétisme de la formation des 'gres à Voltzia' du Massif des Vosges, Can. J. Earth Sci., 14, 1490-1514, 1977.

Bogdanoff, S. and Schott, J.J., Etude paléomagnétique et analyse tectonique dans les schistes rouges permiens du sud de l'Argentera, Bull. Soc. géol. France, 19, 909-916, 1977.

Duringer, P. & Hagdorn, H., La zonation par cératites du Muschelkalk supérieur lorrain. Diachronisme des faciès et migration vers l'Ouest du dispositif sédimentaire, Bull. Soc. géol. France, 3, 601-609, 1987.

Fabre, A., Le temps dans la construction des courbes de dérive apparente du pôle paléomagnétique, Ph.D. thesis, pp 156, Univ. Brest, France, 1986.

Fisher, R.A., Dispersion on a sphere, Proc. R. Soc. London, Ser. A, 217, 295-305, 1953.

Galbrun, B. and Daly, L., Etablissement d'un pôle paléomagnétique pour la plate-forme européenne stable à l'époque toarcienne, C. R. Acad. Sci. Paris, 305, 773-776, 1987.

Gallet, Y., La Magnétostratigraphie: Aspects fondamentaux et appliqués, Ph.D. thesis, pp 219, Univ. Paris VII, 1988.

Hijab, B.R. and Tarling, D.H., Lower Jurassic paleomagnetic results from Yorkshire, England, and their implications, Earth Planet. Sci. Lett., 60, 147-154, 1982.

Lowrie, W., Identification of ferromagnetic minerals in a rock by coercivity and unblocking temperature properties, Geoph. Res. Let., 17, 159-162, 1990.

Marcoux, J., Histoire et Topologie de la NeoTethys, Ph.D. thesis, pp 569, Univ. Paris VII, Paris, France, 1987.

McFadden, B. and McElhinny M., The combined analysis of remagnetization circles and direct observations in palaeomagnetism, Earth Planet. Sci. Lett., 87, 161-172, 1988.

McFadden, B. and McElhinny M., Classification of the reversal test in palaeomagnetism, Geophys. J. Int., 103, 725-729, 1990.

Merabet, N and Daly, L., Détermination d'un pôle paléomagnétique et mise en évidence d'aimantations à polarité normale sur les formations du Permien supérieur du Massif des Maures, Earth Planet. Sci. Lett., 80, 156-166, 1986.

Molina-Garza, R.S., Geissman, J.W., Van der Voo, R., Lucas, S.G. and Hayden, S.N., Paleomagnetism of the Moenkopi and Chinle formations in Central New Mexico: implications for the North American apparent polar wander path and triassic magnetostratigraphy, J. Geophys. Res., 96, 14239-14262, 1991.

Mulder, F.G., Paleomagnetic results from Irish Carboniferous and Triassic rocks, Sci. Proc. R. Dublin Soc., Ser. A, 4, 343-349, 1972.

Rother, K., Gesteins-und paleomagnetische Untersuchungen an Gesteinsproben vom Territorium der DDR aus dem Präkambrium bis zum Tertiär und Folgerungen für die Veränderungen des geomagnetischen Hauptfeldes sowie für geologisch-geotektonische Interpretationsmöglichkeiten Ph.D. thesis, 93 pp, Univ. Berlin, Germany (G.D.R.), 1971.

Tucholka, P. and Gallet, Y., Magnétostratigraphie en forage continental: exemple de la séquence permienne du forage de Couy, Bull. Soc. géol. France, 162, 5, 971-975, 1991.

Turner, P. and Ixer, R.A., Diagenetic development of unstable and stable magnetization in the St Bees Sandstone of northern England, Earth Planet. Sci. Lett., 34, 113-124, 1977.

Van den Ende, C., Paleomagnetism of Permian red beds of the Dome du Barrot, Ph.D. thesis, 171 pp, Univ. Utrecht, Netherlands, 1977.

Van der Voo, R., Phanerozoic paleomagnetic poles from Europe and North America and comparisons with continental reconstructions, Rev. Geophys., 28, 2, 167-206, 1990.

Westphal, M., Bazhenov, M.L., Lauer, J.P., Pechersky, D.M. & Sibuet, J.C., Paleomagnetic implications on the evolution of the Tethys belt from the Atlantic ocean to the Pamirs since the Triassic, Tectonophysics, 123, 37-82, 1986.

Zijderveld, J.D.A., A.C. demagnetization of rocks: analysis of results, in: Methods in Paleomagnetism, D.W. Collinson, K.M. Creer and S.K. Runcorn, eds., 254-286, Elsevier, Amsterdam, 1967.

Zijderveld, J.D.A., Paleomagnetism of Esterel rocks, Ph.D. thesis, 199 pp, Univ. Utrecht, Netherlands, 1975.

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